F70LA Life Insurance Mathematics A (2024/25)
Tutorial 9

1. (a) Consider a contract which provides £1 000 at the moment of death of a life aged 60 in return for a premium of £47 per year payable continuously for life.

   On the basis of the AM92 select survival model with an effective rate of interest of 4% per year, and using the claims acceleration approximation, calculate the (i) expected value and (ii) standard deviation of the present value of the office's profit on the contract.

   (b) Suppose that the office issues a policy of the above type to each of 500 men aged 60. For this block of business, calculate the (i) expected value and (ii) standard deviation of the total present value of the profit.

2. Calculate the values of the following functions on the basis of an effective interest rate of 4% per year and the AM92 survival model, stating any assumptions you use:

   (a) $\bar{A}_{40:\overline{20}|}$.

   (b) ${}_{10|}\ddot{a}_{40:\overline{20}|}$.

   (c) $(IA)_{[40]:\overline{20}|}$.

3. A man aged 45 buys a 20-year endowment insurance with level annual premiums payable throughout the duration of the policy and sum insured payable at the end of the year of death or at maturity. The premium basis is as follows.

Survival model:	AM92 select.
Interest:	4% per annum effective.
Expenses:	10% of the first annual premium and 2% of each subsequent premium with further initial expenses of £50 and renewal expenses (incurred at the time of payment of the second and each subsequent premium) of £4.

   Calculate the annual premium for a sum insured of

   (a) £5 000.

   (b) £10 000.

4. A man aged 30 purchases by a single premium payment, a policy which provides an annuity of £1 000 per year, payable annually in advance from age 65. In the event of death before age 65, a fraction λ of the premium is returned at the end of the year of death.

   (a) On the basis of AM92 ultimate mortality and 4% per year effective interest rate, calculate the net single premium when (i) $\lambda = 0$, (ii) $\lambda = 0.5$ and (iii) $\lambda = 1$.

   (b) Comment briefly on your answer to part (a).

   (c) Now suppose that the annuity is guaranteed to be paid for at least 5 years if the man survives to age 65. In other words, if he survives to age 65, five annual payments of £1000 will be made, with the first on his 65th birthday, whether or not he remains alive to age 69; annuity payments from age 70 onwards will be made *only* if he is alive at the time each payment is due.

   Calculate the premium for each of the three values of λ listed above.

5. A life office sells annuity policies to men aged exactly 60. Each policyholder pays a single premium, denoted by P, and then receives an annuity of £20 000 per year payable annually in arrears (so that the first payment is on the 61st birthday). The office uses the following assumptions to calculate the premium:

Survival model:	AM92 select.
Interest:	6% per annum effective.
Expenses:	Nil.

   (a) Calculate P using the equivalence premium principle.

   (b) Using the value of P calculated in part (a), calculate the probability that, for a single policy, the present value of the profit to the office will be positive.

   (c) Calculate the standard deviation of the present value of the profit on a single policy.

   (d) Now suppose the office sells 1 000 of these policies simultaneously to independent lives, i.e. lives whose future lifetimes are modelled by independent random variables.

   Calculate the value of P such that the probability that the present value of the profit to the office is positive is (ii) 95%, and (ii) 99%.

   (e) Comment briefly on your answers to parts (a) and (d).

Numerical solutions: Q1(a)(i) 178.07; Q1(a)(ii) 377.52; Q1(b)(i) £89 035; Q1(b)(ii) £8 441.60; Q2(a) 0.46491; Q2(b) 8.99821; Q2(c) 9.03788; Q3(a) £182.45; Q3(b) £357.38; Q4(a)(i) £2 764.91; Q4(a)(ii) £2 823.12; Q4(a)(iii) £2 883.84; Q4(c)(i) £2 796.38; Q4(c)(ii) £2 855.26; Q4(c)(iii) £2 916.67; Q5(a) £218 380; Q5(b) 0.39332; Q5(c) £644 409.86; Q5(d)(i) £221 730.37; Q5(d)(ii) £223 118.26.